

Extending & Facilitating Informal Care Networks for People with Epilepsy & Caregivers

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ABSTRACT

About 53 million adults in the U.S. have provided care to an adult or child who needs assistance, and the number of informal caregivers is increasing. To reduce their physical and psychological burdens from long-term care, it is necessary to get support from other people. Focusing on people with epilepsy (PWEs) and their primary informal caregivers (PWECS), my doctoral research aims (1) to understand the needs and challenges in seeking and managing information and support; (2) to explore and design technologies that could increase epilepsy awareness and knowledge so that the potential caregivers can provide proper help; and ultimately, (3) to find a way to extend and facilitate informal care networks so that PWEs and PWECS could reduce the burdens and improve their well-being and health. Focusing on epilepsy awareness and attitudes among stakeholders in elementary schools, I propose studies that will utilize interview, survey, and participatory design methods. My dissertation work will shed a new light on informal care networks that involve secondary caregivers and bystanders as temporary caregivers.

CCS CONCEPTS

• **Applied computing** → **Consumer health**; **Health care information systems**; **Health informatics**; • **Human-centered computing** → *Human computer interaction (HCI)*; *Collaborative and social computing systems and tools*; *Collaborative content creation*.

KEYWORDS

Informal Caregiver, Care Network, Epilepsy, Information Sharing, Awareness

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1 BACKGROUND

More than 20% of Americans have provided care to an adult or child who needs assistance from others, and the number of informal caregivers is increasing [1]. Primary informal caregivers need to perform most of caring work with no financial compensation and oftentimes no training, experiencing burden of care [1]. Family members or friends can support and help as primary informal caregivers [4] but being an informal caregiver comes with its own burden. Long-term primary informal caregivers, for instance, are more likely to develop physical, psychological, socioeconomic problem (e.g., depression, stress, burnout, reducing work hours, giving up working entirely, etc.) that could lead to negative impacts on the well-being of their care recipients and other family members [1, 17]. One way to help the primary informal caregivers with their care burden is to provide social support.

Little research has discussed the ways to build informal care networks that consider the well-beings of both the primary informal caregivers and their care recipients. The focus has instead been more on care recipients, and it might have unintentionally disregarded factors that could negatively impact on primary informal caregivers. Technological solutions for better care suggested by the literature actually add to the primary caregivers' burdens by encouraging them to do more work for care recipients [5].

The focus of my doctoral research is the informal care network surrounding PWEs and PWECS, populations that have been stigmatized by the society and neglected in the HCI/CSCW research. Epilepsy is one of the most common neurological chronic diseases that causes recurrent and unprovoked seizures [9]. It is estimated that there are over three million PWEs in the U.S., and in 2013, about 470,000 children in the U.S. are expected to have active epilepsy [9, 19, 20]. PWEs and PWECS face several challenges that include physical and emotional symptoms, medication side effects, cognitive and memory deficits, care coordination difficulties, risks of sudden unexpected death in epilepsy (SUDEP), and economic burdens [7, 13]. In addition to one's health condition, stigma and social isolation around PWEs and PWECS threaten their well-being and health [6] as well. A seizure in an uncontrolled environment with no access to help from others is especially dangerous with possibly grave consequences. For example, a seizure in water could lead to a submersion injury and drowning [2]. However, due to the stigma and misconceptions, it is difficult for PWEs and PWECS to get proper help from other people. Moreover, parents of children with epilepsy feel more concerned when the children are at school, as it cannot always be guaranteed that other people in the proximity could offer appropriate treatment in case of a seizure [15].

Few studies have focused on designing technologies that could support PWEs and PWECS to receive proper help from other people.

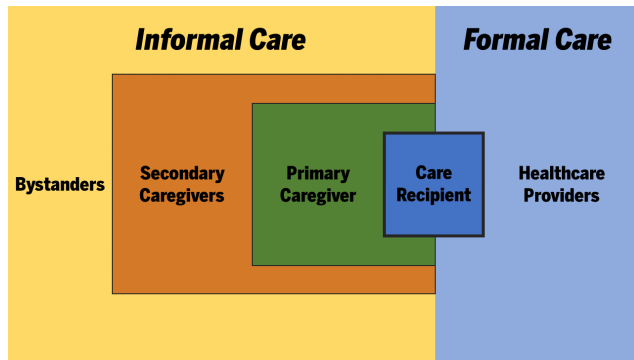


Figure 1: Care Ecosystem

Prior research on epilepsy and seizures has mostly focused on epidemiological aspects of epilepsy [9, 20] as well as burdens placed on PWEs and their caregivers [11, 16]. More recently, research in the HCI field has focused on information and communication technologies (ICTs) that could detect and monitor seizures [3, 10]. However, there is a lack of studies on how to design ICTs that could improve the experiences of PWEs and PWECs when they share information and cooperate with others so that they can reduce their psychological and emotional burdens caused by the risks of having unexpected seizures and stigma from other people.

In my study, I define and focus on three main informal caregiver groups who can provide care (See Figure 1). A **primary informal caregiver**, such as a parent, has the responsibility to take care of a PWE who is not independent. Managing epilepsy and related information of their care recipients is one of their main responsibilities. A **secondary caregiver** personally knows a PWE and can provide occasional care. Compared to primary caregivers, secondary caregivers possess less knowledge of epilepsy and less responsibility for taking care of the PWE. **Bystanders** are any individuals around PWEs in public areas who are not their caregivers. They do not know PWEs and have no responsibilities of taking care of PWEs.

2 RESEARCH QUESTIONS

My doctoral study aims (1) to understand PWEs and PWECs' needs and challenges in information and support management; (2) to explore and design technologies that could increase epilepsy awareness and knowledge and reduce stigma so that PWEs can get proper help from potential temporary caregivers; and ultimately, (3) to find ways to extend and facilitate informal care networks to reduce the burdens of PWEs and PWECs and to improve their well-being and health. I address the following research questions:

- **R.Q.1.** How do PWEs and PWECs seek and manage information and support?
 - (a) What are the needs and challenges they have experienced when they seek, share, and manage information?
 - (b) What are the needs and challenges they have experienced when they look for support?
 - (c) How could ICTs be designed to help them improve their information and support management?
- **R.Q.2.** How do PWEs get proper care when a seizure occurs in their daily lives?

- (a) What challenges do PWEs and PWECs experience in cases where they or their care recipients have seizures when the caregivers are not present?
- (b) What are their coping strategies for handling seizures in different environments?
- (c) How could ICTs be designed to help them inform other people around PWEs so that they can get appropriate and timely care when PWEs are having a seizure?
- **R.Q.3.** How could secondary caregivers in elementary schools provide proper care to children with epilepsy?
 - (a) How do they teach their children/students about how to care for people with diseases?
 - (b) What current knowledge, attitude, and perceptions of epilepsy do teachers and parents of other students have?
 - (c) How could ICTs be designed to improve knowledge and reduce misconceptions and stigma?

3 COMPLETED STUDIES

I finished a focus group study (RQ1) on PWEs and PWECs, which is currently under review for CSCW. This study aimed to explore how PWEs and PWECs seek and manage information and support, along with what needs and challenges they have. Another study (RQ2) has been published in the issue CSCW1 of the Proceedings of the ACM on Human Computer Interaction (PACM HCI) in 2021 [14]. It focuses on how PWEs or PWECs could share seizure first aid information to get help from other people in different places (See Figure 2). I found that social support and temporary or secondary care could be beneficial for the well-being of both primary informal caregivers and their care recipients.

4 WORK IN PROGRESS

I am working on the rest of my doctoral research that will focus on the context of elementary school students and teachers who may potentially become temporary secondary caregivers. I propose research on designing ICTs to facilitate epilepsy care networks in elementary schools in the U.S. As parents of children with epilepsy cannot accompany their children when they are at school, potential caregivers, such as secondary caregivers (teachers or friends who know the student with epilepsy) and bystanders (other students or school staff members who do not know the student with epilepsy), might need to provide temporary care to the students with epilepsy at the school. To encourage them to help PWEs in an appropriate way, their epilepsy awareness should be promoted as well as the stigma should be reduced. In elementary schools, students are more likely to develop proper perceptions of and attitudes toward people with special needs with early education [12]. To do so, the roles of teachers and parents of the students are crucial, as they exert a considerable amount of influence on the students' development [8, 18]. By promoting increased awareness not only at the site of potential caregiving but also at the outskirts of it where the potential secondary caregivers spend most of their time, my hope is to build a more robust network of informal care that can fully assist PWEs physically and mitigate PWECs' concerns.

I propose two studies (See Figure 3). The study (A) will be an exploratory study to investigate how parents and teachers educate

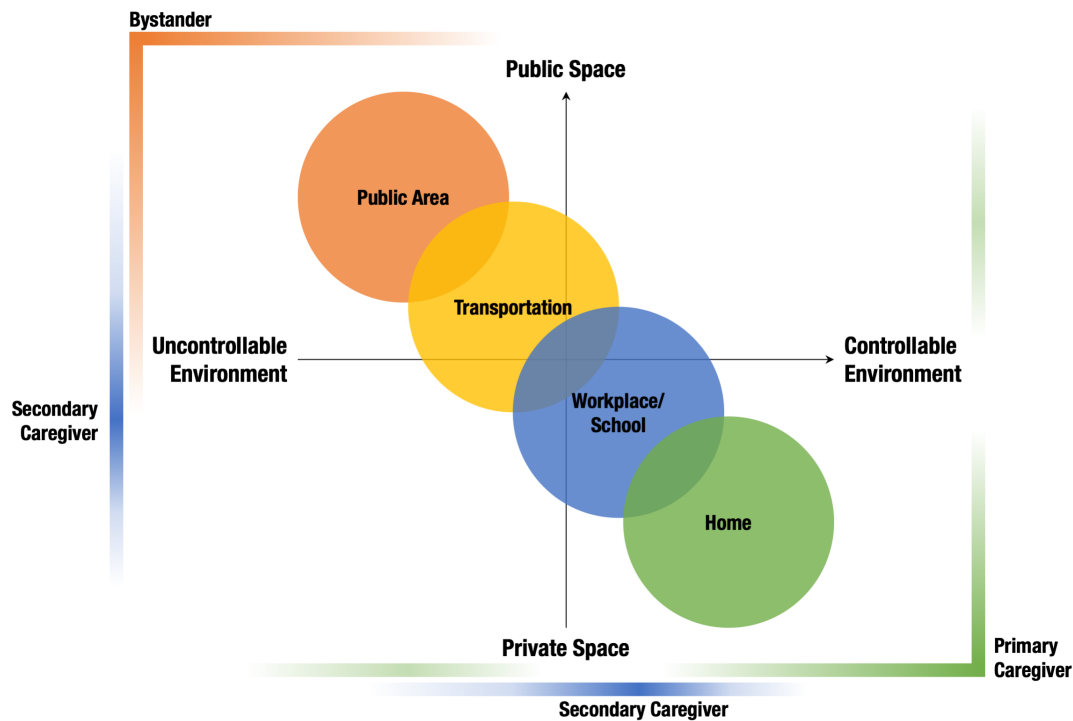


Figure 2: Seizure First Aid Care Framework. The color gradient around the edges indicates the general availability of each caregiver group in each place, with solid color indicating higher availability. [14]

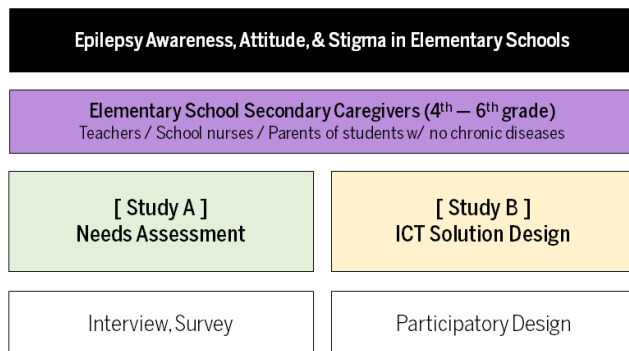


Figure 3: Proposed Studies

their children/students to care for people with diseases or disabilities (RQ3-a) and to identify their current perception of epilepsy encompassing their awareness, attitude, and stigma (RQ3-b). The study (B) will be a participatory design study to design and propose ICTs that work to reduce stigma and increase epilepsy awareness of potential secondary caregivers in elementary schools (RQ3-c).

5 EXPECTED CONTRIBUTIONS

My doctoral research will contribute to epilepsy research and the HCI/CSCW fields. It sheds a new light on informal care networks for both PWEs and PWECs by involving secondary caregivers and

bystanders into the care network. Also, this research identifies their needs and complex challenges in information and support management, as well as the areas where they experience different issues of having seizures and getting proper care. Considering different contexts, it provides frameworks and design implications of future technologies that could increase epilepsy awareness and mitigate stigma among people to effectively involve them as temporary caregivers. The frameworks of informal care networks and design in this work could be extended to provide insights on several additional topics such as other understudied chronic diseases that cause unpredictable symptoms; the need to have access to first aid assistance; a lack of awareness from the general public; and stigma-related issues and concerns.

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